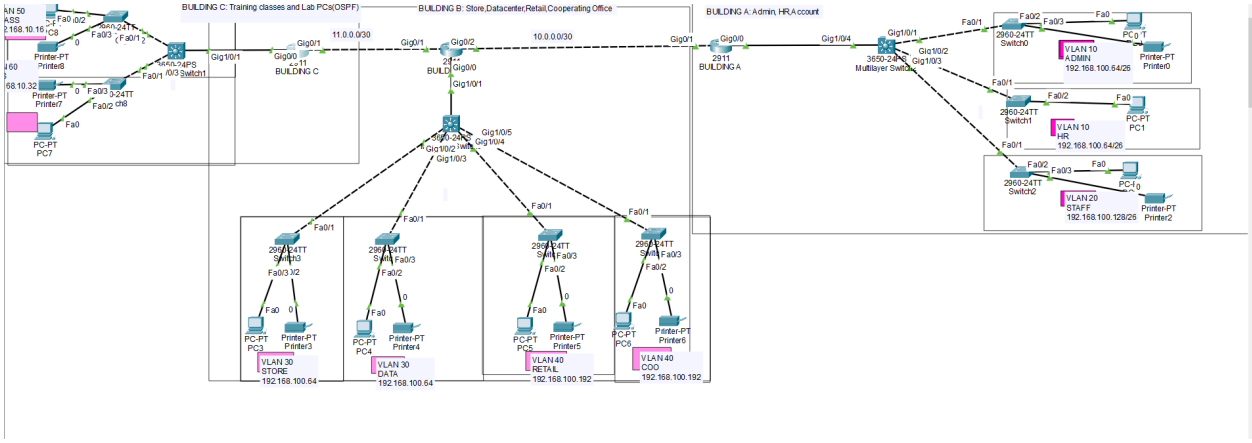
PROJECT DOCUMENTATION:NETWORK DESIGN AND IMPLEMENTATION USING CISCO PACKET TRACER

Network Topology:

The topology consists of a 3 routers connected to 3 multilayer switches and a branch of office network



DEVICES USED:

3x Routers(2911), 3x Multilayer switches, 3x Switches(2960), 9x Pcs and Printers.

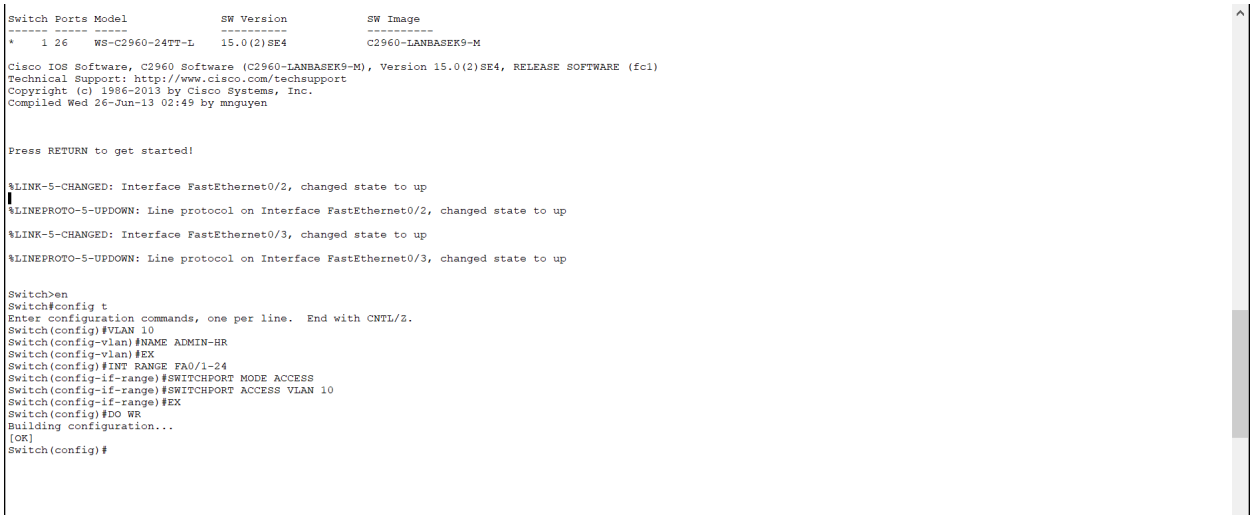
IP ADDRESSING SCHEMES:

Department/ VLANS	Network Address	Subnet Mask	Default Gateway	VLAN ID
ADMIN-HR	192.168.100.64	255.255.255.192	192.168.100.65	VLAN 10
STAFF	192.168.100.128	255.255.255.192	192.168.100.129	VLAN 20
STORE-DATA	192.168.100.64	255.255.255.192	192.168.100.65	VLAN 30
RETAIL-COO	192.168.100.192	255.255.255.192	192.168.100.193	VLAN 40
CLASS	192.168.10.16	255.255.255.240	192.168.10.17	VLAN 50

BASIC SWITCH CONFIGURATION:

I started by configuring the switches with a host name for easy identification. To segment the network and improve security, I created separate VLANs for each department. This prevents broadcast traffic from one department from affecting others.

I also assigned each switchport to its respective VLANs based on the department. For example ports connected to STAFF PCs were assigned to VLAN 20.



```
Switch  Ports  Model              SW Version            SW Image
------  -----  -----              ----------            ----------
*    1  26     WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 26-Jun-13 02:49 by mnguyen


Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

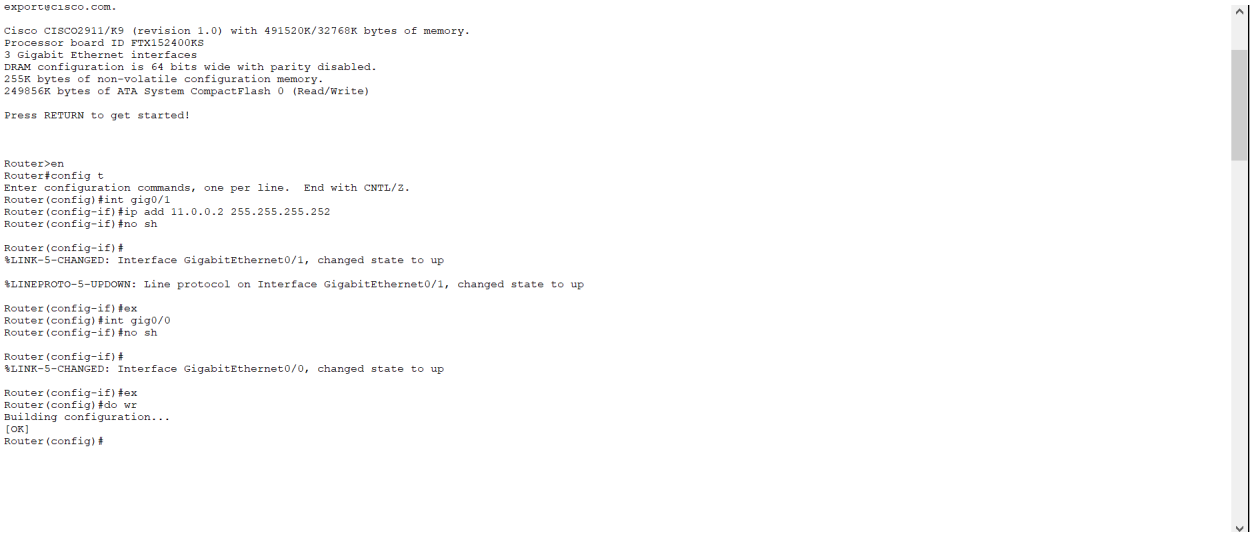
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up


Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#VLAN 10
Switch(config-vlan)#NAME ADMIN-HR
Switch(config-vlan)#EX
Switch(config)#INT RANGE FA0/1-24
Switch(config-if-range)#SWITCHPORT MODE ACCESS
Switch(config-if-range)#SWITCHPORT ACCESS VLAN 10
Switch(config-if-range)#EX
Switch(config)#DO WR
Building configuration...
[OK]
Switch(config)#
```

BASIC ROUTER CONFIGURATION:

I configured the routers by turning on the router interfaces and assigning IP addresses to them making sure they do not shutdown .



```
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/1
Router(config-if)#ip add 11.0.0.2 255.255.255.252
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#ex
Router(config)#int gig0/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

BASIC MULTILAYER SWITCH CONFIGURATION:

I ensured that each multilayer switchport is assigned to its respective VLANs based on the department in each building. For example, in building B, int gig1/0/2 was assigned to VLAN 30

TRUNK PORT CONFIGURATION:

I configured trunk links between multilayer switches and the routers to allow multiple VLAN traffics to pass through a single link using 802.1Q encapsulation.

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/5 (1), with Switch FastEthernet0/1 (40).

Switch(config-if)#int gig1/0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#ex
Switch(config)#int gig1/0/2
Switch(config-if)#ex
Switch(config)#int gig1/0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#ex
Switch(config)#int gig1/0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/4 (1), with Switch FastEthernet0/1 (40).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet1/0/5 (1), with Switch FastEthernet0/1 (40).

% Access VLAN does not exist. Creating vlan 40
Switch(config-if)#ex
Switch(config)#int gig1/0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#ex
Switch(config)#int gig1/0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up
```

INTER-VLAN ROUTING(ROUTER-ON-A-STICK):

Since devices in different VLANs can not communicate by default, I configured sub-interfaces on the router to route traffic between VLANs. Each sub-interfaces serves as the default gateway for its VLAN.

```
Router>EN
Router#CONFIG T
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot1q 30
Router(config-subif)#ip address 192.168.100.65 255.255.255.192
Router(config-subif)#ex
Router(config)#int gig0/0.40
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.40, changed state to up

Router(config-subif)#encapsulation dot1q 40
Router(config-subif)#ip address 192.168.100.193 255.255.255.192
Router(config-subif)#ex
Router(config)#
```

ROUTING PROTOCOL CONFIGURATION:

To enable communication between the buildings and the branch offices, I configured OSPF routing. This allows routers to share network information dynamically.

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 11.0.0.0 0.255.255.255 area 0
Router(config-router)#network 192area 0
02:04:13: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.100.193 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
.                                ^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.10.16 0.0.0.255  area 0
Router(config-router)#network 192.168.10.32 0.0.0.255  area 0
Router(config-router)#ex
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

DHCP CONFIGURATION:

Instead of statically assigning IP addresses to all PCs, I configured DHCP on each router to automatically assign IP addresses, default gateways and DNS to clients.

```
Router(config)#
Router(config)#service dhcp
Router(config)#ip dhcp pool CLASS-POOL
Router(dhcp-config)#network 192.168.10.16 255.255.255.240
Router(dhcp-config)#default-router 192.168.10.17
Router(dhcp-config)#dns 8.8.8.8
Router(dhcp-config)#ex
Router(config)#ip dhcp pool LABS-POOL
Router(dhcp-config)#network 192.168.10.32 255.255.255.240
Router(dhcp-config)#default-router 192.168.10.33
Router(dhcp-config)#dns 8.8.8.8
Router(dhcp-config)#ex
```

CONNECTIVITY TESTING:

Finally, I tested the network using ping and traceroute commands to verify full connectivity between departments, buildings and offices.

RESULT:

Inter-VLAN communication failed. Devices in both VLANs were getting IP conflicts and the router could not differentiate between networks.

WHAT I LEARNED:

Each VLAN represents a separate broadcast domain and must have its own unique subnet. Two different VLANs cannot share the same subnet because the router uses the subnet to identify where to send traffic. When subnets overlap, the routing table is inconsistent and ping fails.

HOW I FIXED IT :

I re-designed the IP addressing scheme to ensure each VLAN had a distinct subnet

Department/ VLANs	Network Address	Subnet Mask	Default Gateway	VLAN ID
ADMIN-HR	192.168.100.64	255.255.255.192	192.168.100.65	VLAN 10
STORE-DATA	192.168.100.0	255.255.255.192	192.168.100.1	VLAN 30

After the IP scheme and updating the sub-interfaces on the router, connectivity was restored and all pings were successful.

This taught me the importance of proper IP planning before configuration.